

# Parametric, Polar and Vector-Valued Calculus — Error Analysis

## Answer Key

### AP Calculus BC

#### Quick Answers

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|-----------------------------------------|-------------------------------------|
| 1. $-\frac{1}{4}$                       | 5. 0, -1, 1                         |
| 2. 12.04                                | 6. $\frac{2\pi}{3}, \frac{4\pi}{3}$ |
| 3. $\frac{4\pi}{3}$                     | 7. 1.32                             |
| 4. $-\frac{2}{3} + \frac{4\sqrt{2}}{3}$ | 8. —                                |
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#### Common wrong answers

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1. If they got -4: inverted the quotient and computed  $dx$  by  $dy$  instead of  $dy$  by  $dx$
  2. If they got 13: added the two component velocities instead of taking the magnitude of the velocity vector
  3. If they got 25.13: integrated over the full turn from zero to two pi instead of over the single petal's bounds, sweeping the whole rose
  4. If they got 1.667: integrated the two velocity components separately and added the results instead of integrating the speed
  5. If they got 2.25: substituted the  $x$ -coordinate two straight into the parameter slot instead of solving for the  $t$  that reaches the point
  7. If they got -1.87: added the two velocity components and reported the result as a speed, which came out negative
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#### 1. Find and fix: Ana inverts the slope quotient, $x = t^2 + 2t$ , $y = t^3 - 4t$ , at $t = 1$ .

Step 1: name the error. The chain rule gives  $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$  — the parameter cancels only in that order. Ana computed the reciprocal, which is  $\frac{dx}{dy}$ , the slope of the same curve read against the wrong axis.

Step 2: differentiate each coordinate:  $\frac{dx}{dt} = 2t + 2$  and  $\frac{dy}{dt} = 3t^2 - 4$ .

Step 3: at  $t = 1$ :  $\frac{dy}{dt} = 3 - 4 = -1$  and  $\frac{dx}{dt} = 2 + 2 = 4$ , so  $\frac{dy}{dx} = \boxed{-\frac{1}{4}}$ .

*Watch for:*  $-4$ , the reciprocal. A slope and its reciprocal are both plausible-looking numbers, so the only defence is writing the quotient in the right order every time.

**2. Find and fix: Bo's "speed" of 13 for  $x = t^2 - 3t$ ,  $y = t^3$  at  $t = 2$ .**

Step 1: name what Bo computed. Adding  $x'$  and  $y'$  produces the sum of two components — a quantity with no geometric meaning, since the components lie along perpendicular axes. Speed is the *magnitude* of the velocity vector, which needs the Pythagorean combination.

Step 2: differentiate:  $x'(t) = 2t - 3$  and  $y'(t) = 3t^2$ . At  $t = 2$ :  $x'(2) = 4 - 3 = 1$  and  $y'(2) = 3(4) = 12$ , so the velocity vector is  $\langle 1, 12 \rangle$ .

Step 3: speed  $= \sqrt{1^2 + 12^2} = \sqrt{145} = 12.0415\dots$ , so the speed is  $12.04$ .

Step 4: sanity check — the magnitude of a vector is at least as large as its largest component (12) and at most their sum (13). Bo's 13 is the upper bound, reachable only if the motion were along a single line.

*Watch for:* 13, i.e.  $1 + 12$ .

**3. Find and fix: the area of one petal of  $r = 4 \cos(3\theta)$ .**

(a) Step 1: a petal begins and ends where the curve returns to the pole, so solve  $r = 0$  first:  $\cos(3\theta) = 0$  gives  $3\theta = \pm\frac{\pi}{2}$ , hence  $\theta = -\frac{\pi}{6}$  and  $\theta = \frac{\pi}{6}$ . Those two consecutive zeros bound one petal.

(b) Step 2: set up the area integral over exactly that sweep:

$$A = \frac{1}{2} \int_{-\pi/6}^{\pi/6} (4 \cos 3\theta)^2 d\theta = 8 \int_{-\pi/6}^{\pi/6} \cos^2 3\theta d\theta.$$

Step 3: use  $\cos^2 u = \frac{1+\cos 2u}{2}$ :  $8 \int \frac{1+\cos 6\theta}{2} d\theta = 4 \left[ \theta + \frac{\sin 6\theta}{6} \right]_{-\pi/6}^{\pi/6}$ . At both limits  $\sin 6\theta = \sin(\pm\pi) = 0$ , so only the  $\theta$  term survives:  $4 \left( \frac{\pi}{6} + \frac{\pi}{6} \right)$ .

Step 4:

$$A = \frac{4\pi}{3} \approx 4.19$$

Step 5: Cal's limits 0 to  $2\pi$  sweep the entire rose, and for a rose with an odd petal count the full turn traces all three petals *twice* over. His 25.13 is  $8\pi$ , which is six times one petal's area — not an area of any region on the page.

*Watch for:* 25.13.

**4. Find and fix: total distance for  $x = t^2$ ,  $y = \frac{2}{3}t^3$  on  $[0, 1]$ .**

Step 1: name the error. Distance travelled is the accumulation of *speed*,  $\int_0^1 \sqrt{x'^2 + y'^2} dt$ . Integrating the components separately and adding gives the sum of two signed displacements, which is not a length: the square root cannot be moved through a sum.

Step 2:  $x'(t) = 2t$  and  $y'(t) = 2t^2$ , so  $\sqrt{x'^2 + y'^2} = \sqrt{4t^2 + 4t^4} = 2t\sqrt{1 + t^2}$  (positive on  $[0, 1]$ ).

Step 3: substitute  $u = 1 + t^2$ ,  $du = 2t dt$ :

$$\int_0^1 2t\sqrt{1 + t^2} dt = \int_1^2 \sqrt{u} du = \frac{2}{3}u^{3/2} \Big|_1^2.$$

Step 4:

$$\text{distance} = \frac{2(2\sqrt{2} - 1)}{3} \approx 1.22$$

Step 5: why Dev's method fails in one line — his answer would be the same for a particle that reversed direction, because component integrals cancel while speed never does.

*Watch for:* 1.667, from  $\int_0^1 2t dt + \int_0^1 2t^2 dt = 1 + \frac{2}{3}$ .

**5. Find and fix: Elle substitutes  $t = 2$  because  $x = 2$ , on  $x = t^2 + 1$ ,  $y = t^3 - 3t$  at  $(2, -2)$ .**

(a) Step 1: name the error.  $t$  is a parameter, not a coordinate. The only way into the point is to solve  $x(t) = 2$ .

Step 2:  $t^2 + 1 = 2$  gives  $t^2 = 1$ , so  $t = -1$  or  $t = 1$ . Test each against the  $y$ -coordinate:  $y(1) = 1 - 3 = -2$  ✓, while  $y(-1) = -1 + 3 = 2 \neq -2$ . The point  $(2, -2)$  is reached at  $t = 1$  only.

(b) Step 3:  $\frac{dy}{dx} = \frac{3t^2 - 3}{2t}$ , so at  $t = 1$  the numerator is  $3 - 3 = 0$  and the slope is  $0$  — the tangent there is horizontal.

Step 4: note that Elle's  $t = 2$  lands on the point  $(5, 2)$ , a different place on the same curve entirely. Her arithmetic was flawless.

*Watch for:* 2.25, which is  $\frac{3(4) - 3}{4}$  — the slope at the wrong parameter.

**6. Find and fix: where does  $r = 1 + 2 \cos \theta$  reach the pole?**

Step 1: name the error. The pole is where  $r = 0$ , so the equation to solve is  $1 + 2 \cos \theta = 0$ , i.e.  $\cos \theta = -\frac{1}{2}$ . Ravi solved  $\cos \theta = +\frac{1}{2}$ , dropping the sign that the constant term forces.

Step 2: on  $[0, 2\pi)$ ,  $\cos \theta = -\frac{1}{2}$  at the two angles whose reference angle is  $\frac{\pi}{3}$ , in quadrants II and III.

Step 3:

$$\theta = \frac{2\pi}{3} \quad \text{or} \quad \theta = \frac{4\pi}{3}$$

Step 4: check: at  $\theta = \frac{2\pi}{3}$ ,  $\cos \theta = -\frac{1}{2}$  and  $r = 1 + 2(-\frac{1}{2}) = 0$  ✓. These two angles are exactly the limits of the inner loop, which is why this step comes before any area integral for this curve.

*Teaching point:* Ravi's  $\frac{\pi}{3}$  gives  $r = 2$ , a point far from the pole — substituting the answer back refutes it in one line.

### 7. Find and fix: Nia's negative "speed" for $\mathbf{r}(t) = \langle \sin 2t, \cos t \rangle$ at $t = \frac{\pi}{3}$ .

Step 1: the answer is refuted before any arithmetic: speed is a magnitude, and a magnitude is never negative. Adding components can produce a negative number, which is proof the wrong operation was used.

Step 2: differentiate componentwise:  $\mathbf{r}'(t) = \langle 2 \cos 2t, -\sin t \rangle$ . At  $t = \frac{\pi}{3}$ :  $2 \cos \frac{2\pi}{3} = 2(-\frac{1}{2}) = -1$  and  $-\sin \frac{\pi}{3} = -\frac{\sqrt{3}}{2} \approx -0.8660$ .

Step 3: speed  $= |\mathbf{r}'| = \sqrt{(-1)^2 + \left(-\frac{\sqrt{3}}{2}\right)^2} = \sqrt{1 + \frac{3}{4}} = \sqrt{\frac{7}{4}} = 1.3228\dots$ , so the speed is 1.32.

Step 4: check — both components are negative here, so the particle is moving down and to the left, yet its speed is positive. Direction lives in the vector; speed keeps only the size.

*Watch for:*  $-1.87$ , the sum of the components.

### 8. Explain: velocity, speed, slope, and polar limits.

*Open response — flagged for manual review. Model answer below; accept any equivalent reasoning.*

(a) *Velocity* is the vector  $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$ : it carries both how fast and in which direction the particle moves. *Speed* is the scalar  $|\mathbf{v}(t)| = \sqrt{x'(t)^2 + y'(t)^2}$ : the size of that vector, direction discarded, and the integrand for distance travelled. The *slope*  $\frac{dy}{dx} = \frac{y'(t)}{x'(t)}$  is also a scalar, but it describes the *curve*, not the motion — it is unchanged if the particle traverses the same path twice as fast, whereas velocity and speed both double.

(b) A component of velocity is a signed rate along one axis, and the sign records direction:  $x' < 0$  means leftward motion. Speed is  $\sqrt{x'^2 + y'^2}$ , a principal square root of a sum of squares, so it is non-negative by construction; it can only be zero, when the particle is momentarily at rest. A negative "speed" therefore always signals a wrong operation, most often adding components instead of taking the magnitude.

(c) The area formula  $\frac{1}{2} \int r^2 d\theta$  accumulates the region swept by the radius. As  $\theta$  runs from 0 to  $2\pi$ ,  $r = 4 \cos 3\theta$  traces all three petals of the rose, and for an odd petal count it covers each one *twice* (once with  $r > 0$ , once with  $r < 0$ , and  $r^2$  cannot tell the difference). The full-turn integral therefore reports six petal-areas. *Habit:* before integrating, solve  $r = 0$  for the  $\theta$ -values that bound the region, sketch or test one interior angle to confirm the sweep covers the region once, and integrate between two consecutive zeros.

*What a full-credit answer must contain:* vector/scalar labelling of all three quantities, the observation that  $dy/dx$  describes the curve rather than the motion, non-negativity of a magnitude by construction, and a polar-limit habit that begins with solving  $r = 0$ .